

Craig T. Russell, PhD

Senior Machine Learning Scientist, generative and agentic models for biology

London, UK • [linkedin.com/in/ctr26](https://www.linkedin.com/in/ctr26) • github.com/ctr26 • [Google Scholar](#)

Generative and agentic models for biology • LLM post-training • out-of-distribution modelling • virtual cells • open to relocating to San Francisco

SUMMARY

I build generative models that generalise to unseen biology, with the aim of building the first virtual cell. I have spent eight years applying machine learning to biological data, from light-sheet microscopy and terabyte-scale image archives to single-cell transcriptomics and knowledge graphs, and published in *Nature Biotechnology*, *NeurIPS*, and *Nature Methods*. I write the research code and the infrastructure under it, from tools other labs use to agents grounded in real biological databases and training and inference on cloud and HPC.

EXPERIENCE

Senior Machine Learning Scientist, Valence Labs @ Recursion Pharmaceuticals

London, UK • Oct 2024 – present

- Build agents that use real biological databases as tools; they decode within Ensembl, HGNC, and ChEMBL vocabularies, check their tool calls against structured resources, and are grounded by retrieval and verifier ensembles. The gene–gene hypotheses they produce are explainable and draw on several sources.
- Co-authored **TxPert** (*Nature Biotechnology*, 2026), a state-of-the-art transcriptomic perturbation predictor conditioned on multiple biological knowledge graphs; I owned benchmark task design, data curation, and the out-of-distribution ablations.
- Post-train multimodal LLMs over knowledge graphs, literature, and omics, and use reinforcement learning to align agent behaviour with verifiable biological objectives.
- Train large diffusion transformers to generate perturbational biology that generalises to perturbations, cell types, and combinations the model has never seen, rather than interpolating between ones it has.
- Model single-cell and bulk RNA-seq jointly with high-content phenotypic imaging, since each modality pins down what the other leaves open.
- Design active learning strategies that choose the next experiment under a fixed budget, closing the loop between model uncertainty and what the wet lab can run.
- Research engineering lead in a ~30-person research team; I set technical direction and engineering standards across ML and biology, scale training and inference across GPUs in the cloud, and encourage the code review, reproducibility, and shared tooling that carry research prototypes into production.
- Work on evaluation strategy and error analysis for proteome-scale binding-affinity screening.
- Organise the Virtual Cell Journal Club, which brings the ML and wet-lab teams together.

Senior Research Associate and AI Engineering Lead, EMBL-EBI (Uhlmann Group and Bio-Image Archive)

Cambridge, UK • Dec 2022 – Oct 2024

- Supervised 6 PhD students, and set the lab's coding standards, CI, and peer review.
- Created **bioimage_embed** and co-authored **ShapeEmbed** (*NeurIPS* 2025), which learn contour and morphology representations without labels; both ship as production Python used by other labs.
- First author on **bia-binder** (*Bioinformatics* 2025), web-native cloud compute that sits beside the Bio-Image Archive.
- Designed scalable pipelines that process terabytes of microscopy across HPC and cloud, with containerised workflows and automated experiment tracking.

Part-time and consulting roles held alongside EMBL-EBI, 2022 – 2024

AI/ML Founding Engineer (part-time), Amun AI AB

Stockholm, Sweden • 2022 – 2024

- Built a Kubernetes (GKE) model-serving platform on NVIDIA Triton and KServe, with auth, monitoring, and autoscaling, serving 100+ models to 30+ daily users.

AI/ML Engineering Consultant (part-time), DeepMirror

Cambridge and London, UK • 2022 – 2024

- Shipped **MouseMindMapper**, a brain-histology segmentation product that brings in £50k of revenue a year, and owned it end to end, from data and training to packaging and docs.
- Wrote a high-performance C++ cheminformatics fingerprinting library for production.

Data Scientist, Bzoma Group, EMBL-EBI

Cambridge, UK • Dec 2019 – Dec 2023

- Co-authored the successful €5M **AI4LIFE** grant for federated bioimage AI infrastructure, and contributed to its platform architecture and model-sharing strategy.
- Drove large-scale AI microscopy analyses in the Image Data Resource, and worked with Google Cloud on representation learning.
- Taught an annual deep learning course to 40+ researchers, from PhD students to PIs.

Software Engineer (COVID-19 response), European Nucleotide Archive, EMBL-EBI

Cambridge, UK • Mar 2020 – Sept 2020

- Built CI/CD for the **COVID-19 Data Portal**, which let it publish global data daily, and scaled NGS alignment and ETL in Nextflow and Kubernetes as volumes surged.

Computational Microscopist, National Physical Laboratory

London, UK • 2018 – Dec 2019

- Developed new 3D organoid segmentation methods for cancer research, and consulted for MSquared on advanced imaging.

EDUCATION

PhD, Engineering, University of Cambridge • 2014 – 2018 (EPSRC PES-CDT)

Thesis on light-sheet microscopy for tracking particles in large specimens

- Designed and built a new light-sheet microscope with automated acquisition, and wrote algorithms for particle tracking, signal optimisation, and micrometre-scale tomography. Supervised two MRes students and one BSc student.

MRes, Photonics, University of Cambridge and UCL • 2013 – 2014 • structured illumination microscopy reconstruction

MSci, Physics (First-Class Honours), Nottingham Trent University • 2009 – 2013 • top physics graduate

SELECTED PUBLICATIONS

1. **TxPert**. Out-of-distribution transcriptomic perturbation prediction over biological knowledge graphs. *Nature Biotechnology* (2026), co-author. [doi:10.1038/s41587-026-03113-4](https://doi.org/10.1038/s41587-026-03113-4)
2. **ShapeEmbed**. Self-supervised learning of 2D contour representations. *NeurIPS* (2025), second author. [proceedings](https://proceedings.neurips.cc/paper_files/paper/2025/file/ShapeEmbed-main.pdf)
3. **bia-binder**. Web-native cloud compute for bioimage analysis. *Bioinformatics* (2025), *first author*. [doi:10.1093/bioinformatics/btaf412](https://doi.org/10.1093/bioinformatics/btaf412)
4. **MIFA**. Metadata and accessibility standards for reusable AI training datasets in bioimaging. *Nature Methods* (2025), co-author. [doi:10.1038/s41592-025-02835-8](https://doi.org/10.1038/s41592-025-02835-8)
5. **DL4MicEverywhere**. Reproducible, containerised deep learning for microscopy. *Nature Methods* (2024), co-author. [doi:10.1038/s41592-024-02295-6](https://doi.org/10.1038/s41592-024-02295-6)
6. **CIR4MICS**. Synthetic ground truth for benchmarking image-analysis methods. *Bioinformatics* (2023), co-author. [doi:10.1093/bioinformatics/btad587](https://doi.org/10.1093/bioinformatics/btad587)
7. **The COVID-19 Data Portal**. Rapid open data sharing for SARS-CoV-2 research. *Nucleic Acids Research* 49(W1) (2021), co-author. [doi:10.1093/nar/gkab417](https://doi.org/10.1093/nar/gkab417)
8. **mmSIM**. Open toolbox for structured illumination microscopy. *Phil. Trans. R. Soc. A* (2021), *first author*. [doi:10.1098/rsta.2020.0353](https://doi.org/10.1098/rsta.2020.0353)
9. **Frame-localisation OPT**. Reconstruction for optical projection tomography. *Scientific Reports* (2021), *first author*. [doi:10.1038/s41598-021-83454-z](https://doi.org/10.1038/s41598-021-83454-z)

The full list is on [Google Scholar](https://scholar.google.com/citations?user=...).

PATENTS

- **Virtual Cell Foundation Model**, patent pending, 2024 (Recursion)
- **TxPert, transcriptomic perturbation prediction**, patent pending, 2024 (Recursion)

OPEN SOURCE

Author of [bioimage_embed](#); contributor to napari, BioImage Model Zoo, Hypha Platform, DL4MicEverywhere, BIA Binder, and COVID Workflow Manager.

SKILLS

ML and AI. Diffusion transformers, generative modelling, fine-tuning and post-training foundation models, reinforcement learning, agents and tool use, constrained decoding and retrieval grounding, active learning, contrastive and self-supervised learning, OOD and uncertainty, evaluation and benchmark design

Frameworks. PyTorch, Lightning, Hugging Face, Pyro, TensorFlow, scikit-learn

Biology and data. Single-cell and bulk RNA-seq, high-content and phenotypic imaging, histopathology, GNNs, knowledge graphs; UniProt, PDB, Ensembl, NCBI, ChEMBL

Languages. Python (primary), R, C++, Rust, MATLAB, Java

Compute. Multi-GPU (A100, V100), CUDA, distributed training, SLURM, HPC, GCP, AWS

MLOps and infrastructure. Kubernetes, Docker, NVIDIA Triton, KServe, MLflow, CI/CD, Terraform; Nextflow, Snakemake, Airflow

GRANTS, TEACHING, AND SERVICE

- **Grants**. AI4LIFE (2022, €5M EU Horizon, co-author) • EPSRC CDT Studentship (2013–2018, £120k) • Nuffield Research Bursary (2012)
- **Teaching**. Led the Deep Learning for Bioimage Analysis course (2019–2023), 40+ participants a year
- **Supervision**. 6 PhD students in AI and spatial biology, and 3 project students
- **Peer review**. Nature Methods, Scientific Reports, Journal of Microscopy, ISBI (2022, 2023), ICASSP (2024)
- **Talks**. FOM (2018, 2022, 2023), MMC (2018, 2022), CBIAS (2023)